Lecture 1: Introduction & LP Modeling

Week 1: Introduction & LP Modeling

ISE 3200 - Linear Optimization

What is linear optimization?

- Decisions, objective, constraints
- Linearity and divisibility assumptions
- Examples: product mix, diet, blending

Building a model

- 1. Define decision variables with units
- 2. Write the objective
- 3. Write every constraint, including x >= 0

Common pitfalls

- Vague variables ("x = furniture")
- Percentages: x2 >= 0.3(x1 + x2), not x2 >= 0.3
- Forgetting non-negativity